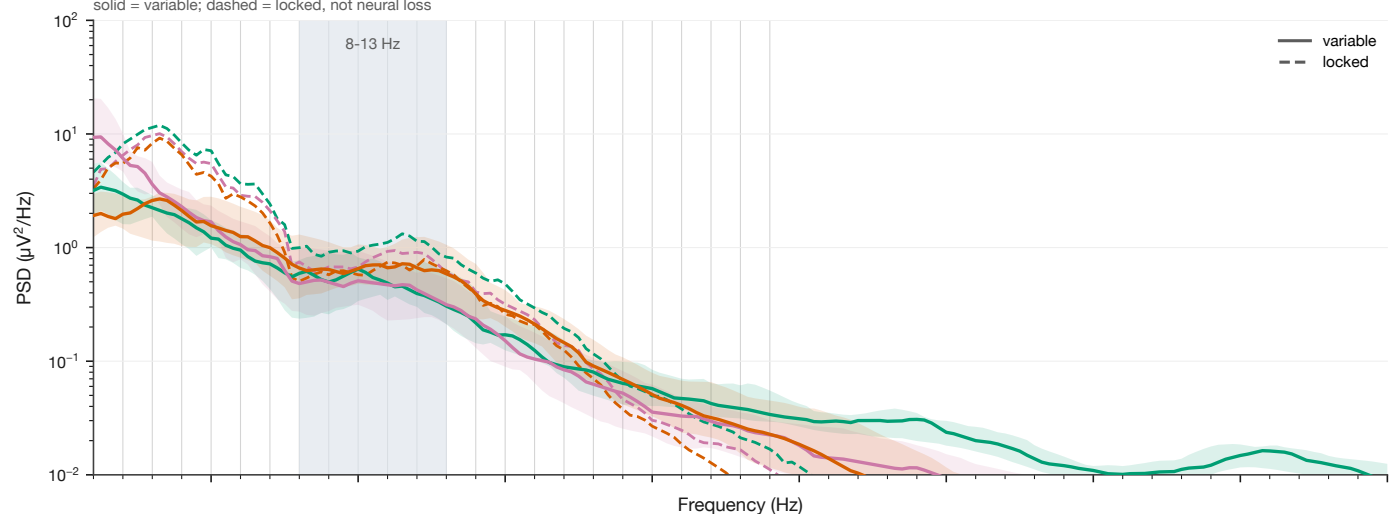


Cohort — 124 paired recordings, 20 participants

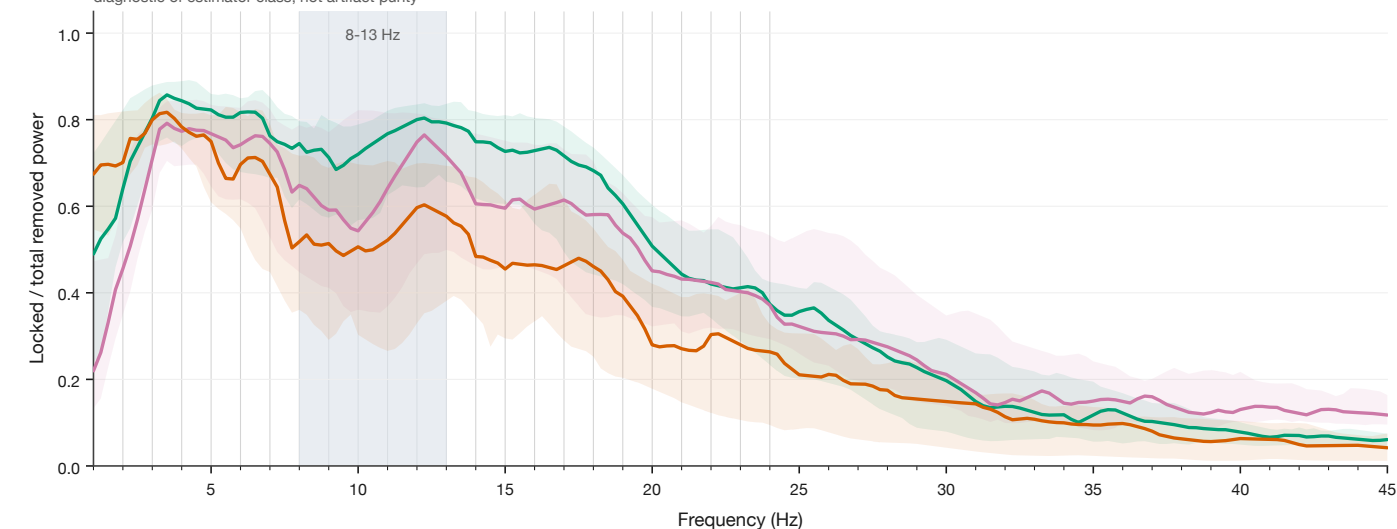
A Removal spectra

solid = variable; dashed = locked, not neural loss



B Removal phase-locking specificity

diagnostic of estimator class, not artifact purity



Descriptive evaluation of saved outputs, not independent validation. All EEG channels; recording median removed. Shaded bands are participant IQR. No signal-preservation or overall winner claim is supported. Spectra of what was subtracted, not of residual EEG. Solid = beat-variable removal; dashed = heartbeat-locked removal. The 8-13 Hz band is both conventional alpha and BCG harmonics 8-13 at typical heart rates; it does not identify neural loss. Frequency minor ticks and the faint comb are 1 Hz (heart rate is not stored on the profile; epoch Welch resolution is ~1 Hz, so harmonics are not resolved as discrete lines). Phase-locking is locked / total removed power: a diagnostic of estimator class, not artifact purity. Blocks below 8 beats are left missing, not replaced. BCGNet outputs include time used to train the subject model (70/15/15 split, then applied to the full recording). AAS and PCA-OBS are not trained that way. Residual panels are not a matched in-sample comparison. Recordings without a usable output are excluded rather than substituted (AAS 16; PCA-OBS 16; BCGNet 11).